

MARY MONROE

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EDUCATION

University of Colorado, Boulder

Aug 2022 - Present (est. May 2027)

Ph.D. candidate in Computer Science (GPA: 4.0/4.0)

Advised by Dr. Bo Waggoner and Dr. Rafael Frongillo

University of Maryland, College Park

Aug 2017 - May 2021

Dual B.S. in Computer Science and Mathematics (GPA: 3.99/4.0)

Member of Honors College, graduated Summa Cum Laude

RESEARCH PUBLICATIONS & PRESENTATIONS

Authors listed alphabetically unless marked with *.

Peer Prediction with More Signals than Reports.

Rafael Frongillo, Ian Kash, and Mary Monroe, *ACM Conference on Economics and Computation (EC)*, 2026. [\[Preprint\]](#)

Equilibria in Large Position-Optimization Games.

Melody Hsu, Mary Monroe, Anish Thilagar, and Rafael Frongillo, *ACM Conference on Economics and Computation (EC)*, 2026. [\[Preprint\]](#)

Hedging and Approximate Truthfulness in Traditional Forecasting Competitions.*

Mary Monroe, Anish Thilagar, Melody Hsu, and Rafael Frongillo, *AAAI Conference on Artificial Intelligence*, 2025.
Oral presentation; acceptance rate: 2.3% (300/12,957). [\[Link\]](#)

Manipulation-Resistant AI Oracles via Credible Neutrality.*

Mary Monroe, Gabriel P. Andrade, Rafael Frongillo, and Bo Waggoner.

Presented at *EC Workshop in Game Theory and Mechanism Design with LLMs*, 2026. [\[Blog Post\]](#)

Robust Forecast Aggregation via Additional Queries.

Rafael Frongillo, Mary Monroe, and Bo Waggoner, *under submission*. [\[Preprint\]](#)

Public Projects with Preferences and Predictions.

Mary Monroe and Bo Waggoner, *under submission*.

Invited poster at *ACM Conference on Economics and Computation (EC)*, 2025. [\[Preprint\]](#)

WORK EXPERIENCE

Gensyn

May 2026 - Aug 2026

PhD Research Intern

- Worked with the Gensyn research team to develop AI oracles for settling blockchain-based prediction markets.
- Theoretically modeled security threats to design mechanisms for eliciting high-quality information on-chain.

Amazon Web Services

Aug 2021 - Aug 2022 (full-time) / June 2020 - Aug 2020 (intern)

Software Development Engineer

Crystal City, VA

- Worked with engineers to create a service leveraging AI that connects multiple SaaS products into one platform.

Undergraduate Research Assistant*University of Maryland*

Aug 2020 - May 2021

College Park, MD

- Worked with Dr. David Mount in computational geometry and Dr. John Dickerson in supervised learning.

Geon Technologies*Signal Processing Engineer Intern*

June 2019 - Aug 2019

*Columbia, MD***U.S. Army Research Lab***Signal Processing Engineer Intern*

June 2018 - Aug 2018

Adelphi, MD

SERVICE & TEACHING**EC Gender Inclusion Workshop Co-Organizer***ACM Conference on Economics and Computation (EC)*

July 2026

Rome, Italy

Co-organized the annual EC Gender Inclusion Workshop, with invited talks, lunches, and discussions.

Teaching Assistant*University of Colorado and University of Maryland*

Sept 2018 - Fall 2025

Led weekly tutorials and graded assignments for Algorithms, Linear Algebra, and Object-Oriented Programming.

Algorithmic Fairness Reading Group Co-Founder/Coordinator*University of Colorado, Boulder*

Aug 2023 - Aug 2024

Boulder, CO

Worked with co-founders to create syllabus of papers, reach out to communities for participation, and plan meetings.

Algorithmic Economics Reading Group Coordinator*University of Colorado, Boulder*

May 2023 - Aug 2024

Boulder, CO

Planned and hosted reading group meetings, scheduled talks, and invited speakers in the area to present research.

Graduate Peer Mentor*University of Colorado, Boulder*

2024 - 2026

Boulder, CO

Advised and supported incoming PhD students; provided feedback to prospective PhD applicants.

Reviewer: EC '26, AAAI '26, '27, EAMMO '25, '26

AWARDS**Outstanding Research Paper Award***University of Colorado Boulder, Computer Science Department*

2025

*Boulder, CO***Presidential Scholarship***University of Maryland*

2017-2021

College Park, MD

SKILLS

Research: Game theory, probability & statistics, learning theory, convex optimization, mechanism design

Languages & libraries: Python, R, Matlab, Java, C, PyTorch, TensorFlow, scikit-learn, cvxpy